

Identifying and evaluating algebraic expressions



1 Match the following terms with their definitions:

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|------------------------|---|
| a Constant | i Symbol that represents an unknown number |
| b Variable | ii A purely numeric term in an algebraic expression |
| c Coefficient | iii The value that indicates how many of a variable in a term |
| d Algebraic term | iv Term including a variable |
| e Algebraic expression | v A collection of variables, numbers, grouping symbols, and operations. |

2 State the numerical coefficient of $-2.58t^4$.

3 Consider the expression $x + 6$.

- a Find the value when $x = 4$.
- b Find the value when $x = 9$.
- c What happens to the value of $x + 6$ as we substitute in different values of x ?

4 Evaluate the following expressions to one decimal place:

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|------------------------------|------------------------------|
| a $14.5 - 4x$ when $x = 4.2$ | b $18.6 - 3x$ when $x = 4.1$ |
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5 If $m = -3$ and $n = 4$, evaluate the following expressions:

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| a $mn - (m - n)$ | b $m^2 + 9n$ |
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6 Evaluate the following expressions:



- a $\frac{m}{2m}$ when $m = 5$
- b $\frac{6}{7x+3}$ when $x = 3$
- c $\frac{a}{11} + \frac{b}{11}$ when $a = 5$ and $b = 2$
- d $\frac{a}{1-r}$ when $a = 3$ and $r = \frac{1}{10}$
- e $\frac{p+4}{q}$ when $p = -8$ and $q = -8$
- f $\frac{3a}{2b-6}$ when $a = 4$ and $b = 14$
- g $\frac{2p+4}{3q+5}$ when $p = 2$ and $q = 4$
- h $\frac{4r-2}{3s-5}$ when $r = 6$ and $s = 7$
- i $\frac{x^2}{3} + \frac{y^3}{2}$ when $x = -4$ and $y = 3$
- j $s(12-t)$ when $s = -3$ and $t = 10$
- k $x(2-4y^2)$ when $x = -2$ and $y = -2$
- l $a(b-13) + b^2$ when $a = -9$ and $b = 10$
- m $3a^2 - 4b^3$ when $a = 4$ and $b = 2$
- n $8p^2 + 5q^3$ when $p = 3$ and $q = -2$
- o $5y - z^2 + 9$ when $y = 7$ and $z = 4$
- p $\frac{1}{2}y^2 - 6z + 1$ when $y = -2$ and $z = \frac{5}{6}$
- q $y^3 + 4yz - 2x^2$ when $x = -4$, $y = 2$ and $z = 1$
- r $(u+v)(w-y)$ when $u = 5$, $v = 8$, $w = 2$ and $y = 10$

7 For $x = 5$ and $y = 4$, evaluate $\sqrt{2x^2 + 4y + 6}$ correct to two decimal places.

Additional questions

8 Write the expression $9x$ in words.



9 State whether the following expressions are like terms.

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|------------------|-------------------|---------------------------|---------------------|
| a $11p$ and $3p$ | b $2p$ and 15 | c $9p$ and $6q$ | d 12 and 8 |
| e $5h$ and $5hk$ | f $4h^4$ and $3h$ | g $8h^3k^4$ and $9k^4h^3$ | h $6h^k$ and $7k^h$ |

10 For the following algebraic expressions:

- i State the number of terms.
- ii Identify the numerical coefficient of the first term.
- iii Identify the constant term.
- iv State whether it contains like terms.

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|----------------------|-----------------|
| a $2x + 4$ | b $7y + 3 + 5x$ |
| c $3x + 2y - 8x + 9$ | d $8p + 5$ |

11 Consider the algebraic expression $3x + 10 + 4x$.

- a State the number of terms.
- b Identify the numerical coefficient of the third term.
- c Identify the constant term.
- d State whether it contains like terms.

12 For each of the following, determine if the statement is true or false. Explain your reasoning.

- a An expression must include a variable.
- b Any expression with more than one term can be simplified.